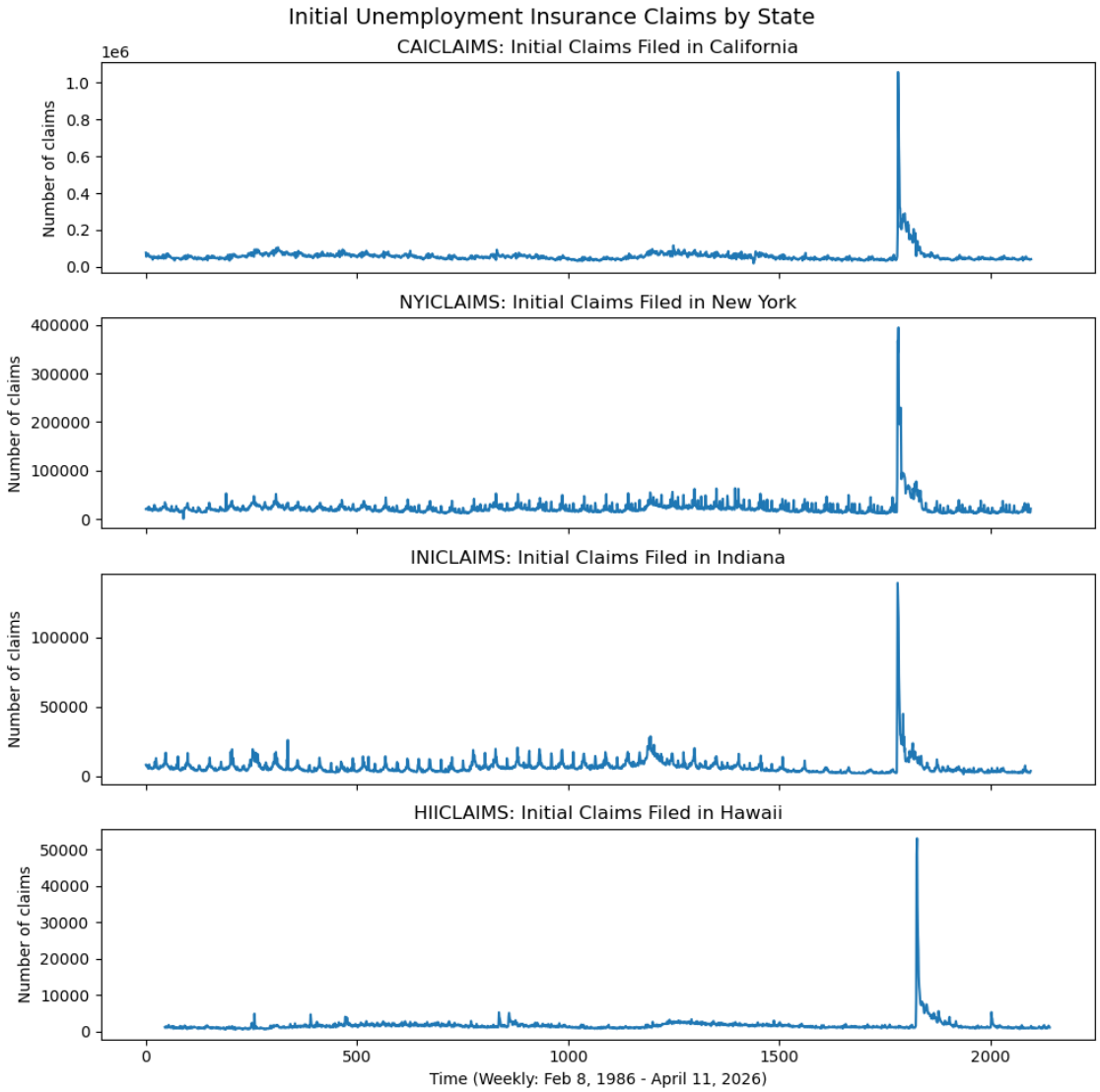
Econ148Proj03

April 28, 2026

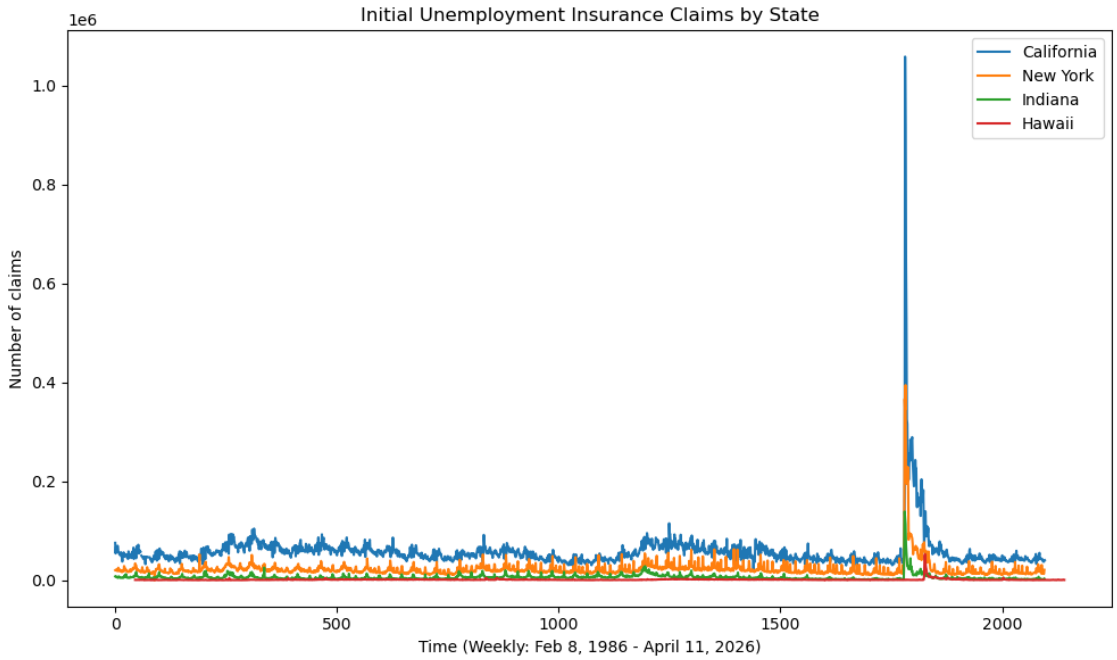
[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import statsmodels.api as sm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX

from sklearn.metrics import mean_squared_error

import os
import itertools
import warnings
from joblib import Parallel, delayed

warnings.filterwarnings("ignore")
```

1 Data Preview

[2]:
```
datasets = ["CAICLAIMS", "NYICLAIMS", "INICLAIMS", "HIICLAIMS"]
state_names = {
    "CAICLAIMS": "California",
    "NYICLAIMS": "New York",
    "INICLAIMS": "Indiana",
    "HIICLAIMS": "Hawaii"
}

claims_dfs = {}

fig, axes = plt.subplots(4, 1, figsize=(10, 10), sharex=True)

for ax, code in zip(axes, datasets):
    df = pd.read_csv(f"Data/{code}.csv")
    df.rename(columns={code: "Claims"}, inplace=True)

    claims_dfs[code] = df
    globals()[code] = df

    ax.plot(df["Claims"])
    ax.set_title(f"{code}: Initial Claims Filed in {state_names[code]}")
    ax.set_ylabel("Number of claims")

axes[-1].set_xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
fig.suptitle("Initial Unemployment Insurance Claims by State", fontsize=14)
plt.tight_layout()
plt.show();
```



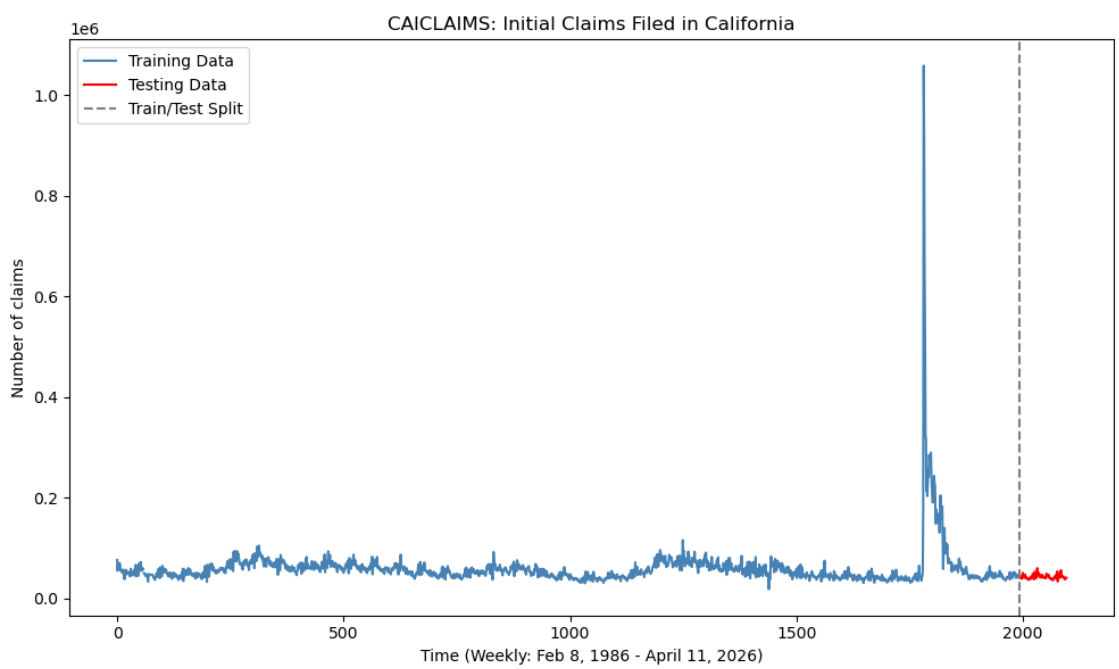
[3]:
```
fig, ax = plt.subplots(figsize=(10, 6))

for code in datasets:
    df = claims_dfs[code]
    ax.plot(df["Claims"], label=state_names[code])

ax.set_xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
ax.set_ylabel("Number of claims")
ax.set_title("Initial Unemployment Insurance Claims by State")
ax.legend()
plt.tight_layout()
plt.show();
```


Initial Unemployment Insurance Claims by State
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Number of claims
0
500
1000
1500
2000
Time (Weekly: Feb 8, 1986 - April 11, 2026)
California
New York
Indiana
Hawaii

2 CAICLAIMS Model Testing

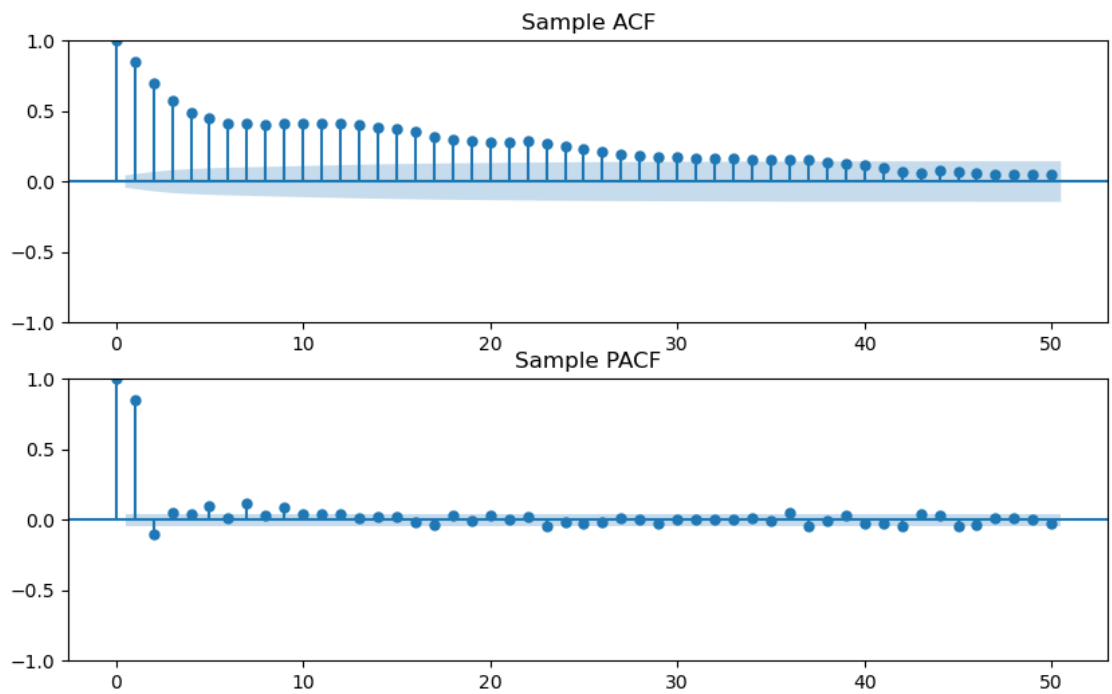
2.1 1. ARIMA Baseline Model

[4]:
```
y_CA = claims_dfs["CAICLAIMS"]["Claims"]

test_horizon = 104  # final 2 years
train_end = len(y_CA) - test_horizon
ttrain = np.arange(train_end)
ttest = np.arange(train_end, len(y_CA))
tpred = np.arange(train_end, len(y_CA) + test_horizon)  # future time points␣
↪for forecasting, 2 years ahead

ytrain_CA = y_CA.iloc[:train_end]
ytest_CA = y_CA.iloc[train_end:]

plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```

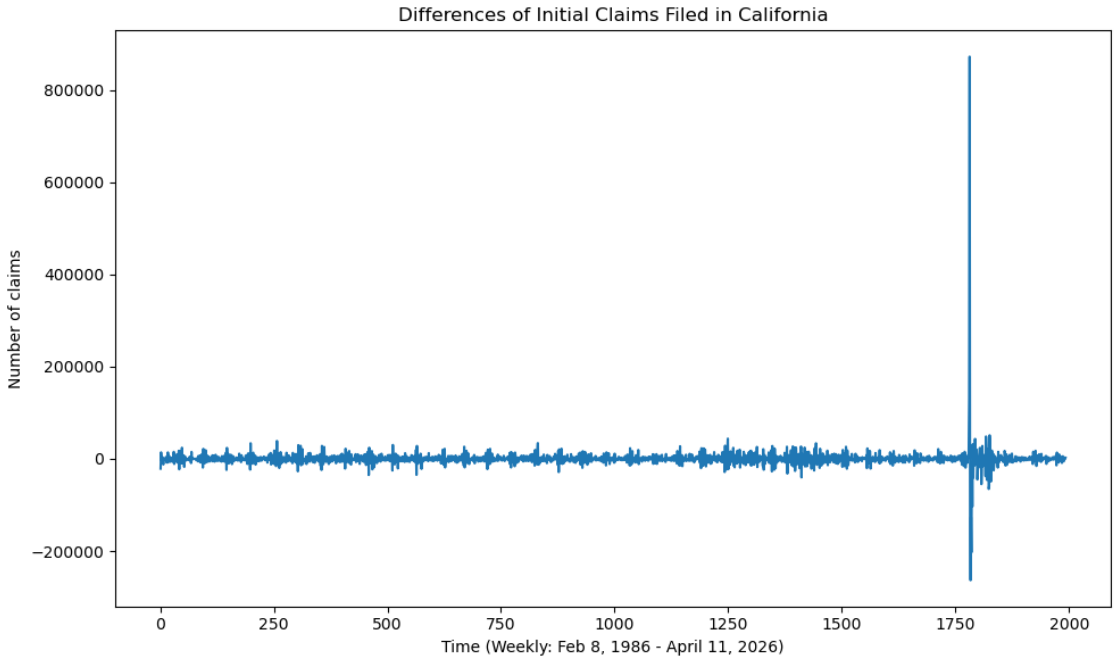


2.1.1 Training Data Time Series Analysis

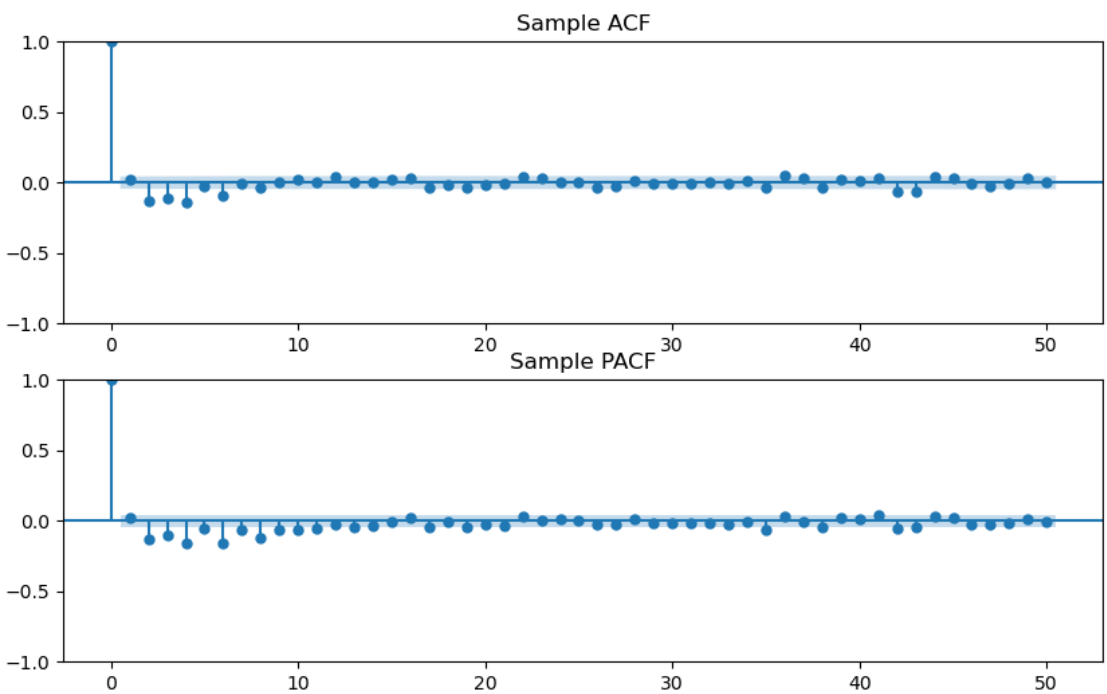
[5]:
```
p_max = 50
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6))
plot_acf(ytrain_CA.dropna(), lags=p_max, ax=ax1, title="Sample ACF")
plot_pacf(ytrain_CA.dropna(), lags=p_max, ax=ax2, title="Sample PACF")
plt.show();
```



[6]:
```
ytrain_CA_diff = ytrain_CA.diff()

plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.title("Differences of Initial Claims Filed in California")
plt.tight_layout()
plt.show();
```



[7]:
```
p_max = 50
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6))
plot_acf(ytrain_CA_diff.dropna(), lags=p_max, ax=ax1, title="Sample ACF")
plot_pacf(ytrain_CA_diff.dropna(), lags=p_max, ax=ax2, title="Sample PACF")
plt.show();
```



2.1.2 Parameter Selection (Grid Search)

[8]:
```
pmax, dmax, qmax = 8, 1, 6

def fit_arima(params):
    warnings.filterwarnings("ignore")
    p, d, q = params

    try:
        model = ARIMA(ytrain_CA, order=(p, d, q)).fit()
        forecast = model.get_forecast(steps=len(ytest_CA))
        yhat = forecast.predicted_mean
        yhat.index = ytest_CA.index
        rmse = np.sqrt(mean_squared_error(ytest_CA, yhat))

        return {
            "p": p,
            "d": d,
            "q": q,
            "AIC": model.aic,
            "BIC": model.bic,
            "RMSE": rmse
        }
    except Exception:
        return None
```

[9]:
```
cache_path_arima = "arima_grid_results_CAICLAIMS.csv"

param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

if os.path.exists(cache_path_arima):
    print(f"Loading cached ARIMA grid search results from {cache_path_arima}")
    results_arima_df = pd.read_csv(cache_path_arima)

else:
    print("Running ARIMA grid search...")

    results_arima = Parallel(n_jobs=-1, verbose=10)(
        delayed(fit_arima)(params) for params in param_grid
    )

    results_arima = [r for r in results_arima if r is not None]
    results_arima_df = pd.DataFrame(results_arima)
```

```
    if results_arima_df.empty:
        raise ValueError("ARIMA grid search failed: no models were successfully␣
↪fit.")

    results_arima_df.to_csv(cache_path_arima, index=False)
    print(f"Saved ARIMA grid search results to {cache_path_arima}")
```

```
Running ARIMA grid search…

[Parallel(n_jobs=-1)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=-1)]: Done   2 tasks      | elapsed:    4.8s
[Parallel(n_jobs=-1)]: Done   9 tasks      | elapsed:    5.5s
[Parallel(n_jobs=-1)]: Done  16 tasks      | elapsed:    6.1s
[Parallel(n_jobs=-1)]: Done  25 tasks      | elapsed:    7.4s
[Parallel(n_jobs=-1)]: Done  34 tasks      | elapsed:    9.6s
[Parallel(n_jobs=-1)]: Done  45 tasks      | elapsed:   13.3s
[Parallel(n_jobs=-1)]: Done  56 tasks      | elapsed:   18.9s
[Parallel(n_jobs=-1)]: Done  69 tasks      | elapsed:   26.0s
[Parallel(n_jobs=-1)]: Done  82 tasks      | elapsed:   33.7s
[Parallel(n_jobs=-1)]: Done  97 tasks      | elapsed:   42.7s
[Parallel(n_jobs=-1)]: Done 124 out of 126 | elapsed:  1.1min remaining:    1.0s

Saved ARIMA grid search results to arima_grid_results_CAICLAIMS.csv

[Parallel(n_jobs=-1)]: Done 126 out of 126 | elapsed:  1.1min finished
```

2.1.3 Best Models by Evaluation Metrics

[10]:
```
best_aic = results_arima_df.loc[results_arima_df["AIC"].idxmin()]
best_bic = results_arima_df.loc[results_arima_df["BIC"].idxmin()]
best_rmse = results_arima_df.loc[results_arima_df["RMSE"].idxmin()]

print(
    f"Best model by AIC: ARIMA({int(best_aic['p'])}, "
    f"{int(best_aic['d'])}, {int(best_aic['q'])}), "
    f"AIC: {best_aic['AIC']:.3f}, ",
    f"BIC: {best_aic['BIC']:.3f}, ",
    f"RMSE: {best_aic['RMSE']:.3f}"
)
print(
    f"Best model by BIC: ARIMA({int(best_bic['p'])}, "
    f"{int(best_bic['d'])}, {int(best_bic['q'])}), "
    f"AIC: {best_bic['AIC']:.3f}, ",
    f"BIC: {best_bic['BIC']:.3f}, ",
    f"RMSE: {best_bic['RMSE']:.3f}"
)
print(
    f"Best model by RMSE: ARIMA({int(best_rmse['p'])}, "
    f"{int(best_rmse['d'])}, {int(best_rmse['q'])}), "
    f"AIC: {best_rmse['AIC']:.3f}, ",
    f"BIC: {best_rmse['BIC']:.3f}, ",
    f"RMSE: {best_rmse['RMSE']:.3f}"
)
```

```
Best model by AIC: ARIMA(3, 1, 5), AIC: 45436.385,  BIC: 45486.757,  RMSE:
4804.857
Best model by BIC: ARIMA(0, 1, 6), AIC: 45437.527,  BIC: 45476.705,  RMSE:
4809.158
Best model by RMSE: ARIMA(8, 1, 0), AIC: 45470.097,  BIC: 45520.469,  RMSE:
4598.933
```

[11]:
```
best_aic_model = ARIMA(ytrain_CA, order=(int(best_aic['p']),␣
 ↪int(best_aic['d']), int(best_aic['q']))).fit()
print(best_aic_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(3, 1, 5)   Log Likelihood              -22709.189
Date:                Tue, 28 Apr 2026   AIC                          45436.378
Time:                        18:01:29   BIC                          45486.750
Sample:                             0   HQIC                         45454.877
                               - 1993
Covariance Type:                  opg
==============================================================================
```

```
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -1.2513      0.063    -19.747      0.000      -1.376      -1.127
ar.L2          0.1546      0.096      1.617      0.106      -0.033       0.342
ar.L3          0.4671      0.053      8.800      0.000       0.363       0.571
ma.L1          1.1900      0.063     18.864      0.000       1.066       1.314
ma.L2         -0.4552      0.095     -4.795      0.000      -0.641      -0.269
ma.L3         -0.8956      0.049    -18.227      0.000      -0.992      -0.799
ma.L4         -0.3152      0.060     -5.250      0.000      -0.433      -0.198
ma.L5         -0.1195      0.029     -4.144      0.000      -0.176      -0.063
sigma2      5.025e+08   7.26e-10   6.92e+17      0.000    5.02e+08    5.02e+08
================================================================================
===
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):
130909306.96
Prob(Q):                              0.89   Prob(JB):
0.00
Heteroskedasticity (H):              19.36   Skew:
31.62
Prob(H) (two-sided):                  0.00   Kurtosis:
1257.28
================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
8.72e+32. Standard errors may be unstable.
```

[12]:
```
best_bic_model = ARIMA(ytrain_CA, order=(int(best_bic['p']),␣
↪int(best_bic['d']), int(best_bic['q']))).fit()
print(best_bic_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(0, 1, 6)   Log Likelihood              -22711.764
Date:                Tue, 28 Apr 2026   AIC                          45437.527
Time:                        18:01:30   BIC                          45476.705
Sample:                             0   HQIC                         45451.916
                               - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1         -0.0642      0.005    -13.086      0.000      -0.074      -0.055
ma.L2         -0.2220      0.012    -17.828      0.000      -0.246      -0.198
```

```
ma.L3         -0.1540      0.029     -5.404      0.000      -0.210      -0.098
ma.L4         -0.1782      0.011    -16.274      0.000      -0.200      -0.157
ma.L5         -0.0327      0.026     -1.250      0.211      -0.084       0.019
ma.L6         -0.0917      0.014     -6.362      0.000      -0.120      -0.063
sigma2      5.024e+08   1.03e-10   4.89e+18      0.000    5.02e+08    5.02e+08
================================================================================
===
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):
131557220.62
Prob(Q):                              0.93   Prob(JB):
0.00
Heteroskedasticity (H):              19.32   Skew:
31.67
Prob(H) (two-sided):                  0.00   Kurtosis:
1260.38
================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
1.17e+33. Standard errors may be unstable.
```

[13]:
```python
best_rmse_model = ARIMA(ytrain_CA, order=(int(best_rmse['p']),␣
↪int(best_rmse['d']), int(best_rmse['q']))).fit()
print(best_rmse_model.summary())
```

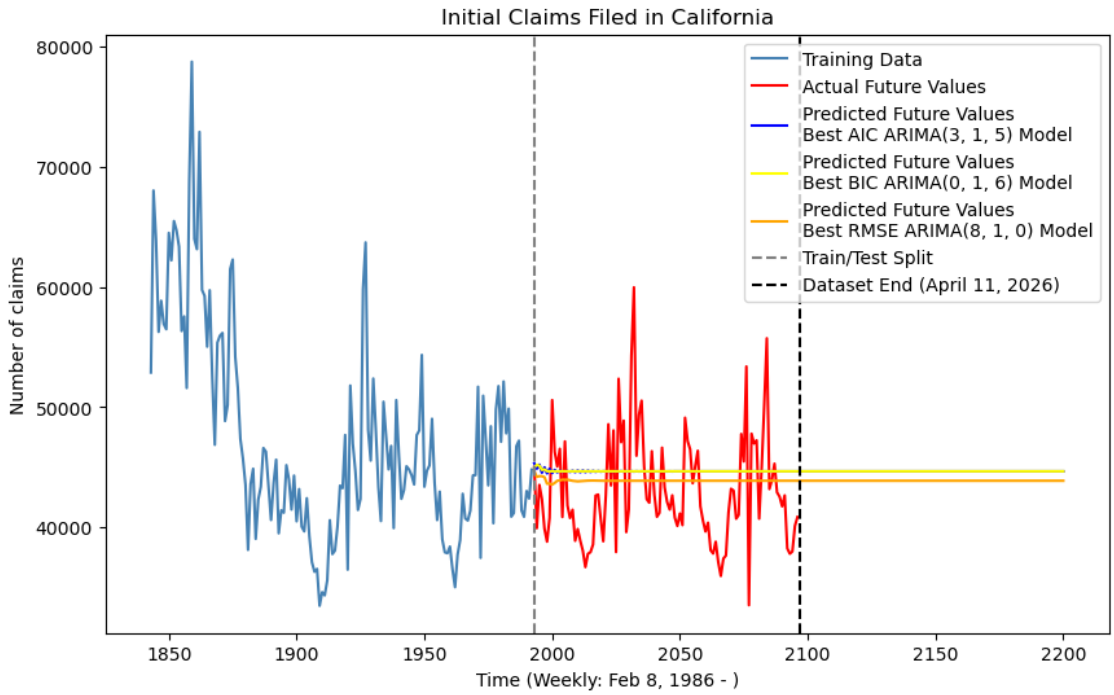
```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(8, 1, 0)   Log Likelihood              -22726.049
Date:                Tue, 28 Apr 2026   AIC                          45470.097
Time:                        18:01:31   BIC                          45520.469
Sample:                             0   HQIC                         45488.597
                               - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.0423      0.004    -10.263      0.000      -0.050      -0.034
ar.L2         -0.2009      0.010    -20.179      0.000      -0.220      -0.181
ar.L3         -0.1483      0.042     -3.528      0.000      -0.231      -0.066
ar.L4         -0.2118      0.009    -22.330      0.000      -0.230      -0.193
ar.L5         -0.0855      0.030     -2.830      0.005      -0.145      -0.026
ar.L6         -0.1808      0.014    -13.215      0.000      -0.208      -0.154
ar.L7         -0.0648      0.031     -2.104      0.035      -0.125      -0.004
ar.L8         -0.1198      0.009    -13.988      0.000      -0.137      -0.103
```

```
sigma2      5.045e+08   2.27e-10   2.22e+18      0.000    5.05e+08    5.05e+08
==============================================================================
===
Ljung-Box (L1) (Q):                   0.05   Jarque-Bera (JB):
120290851.74
Prob(Q):                              0.82   Prob(JB):
0.00
Heteroskedasticity (H):              18.70   Skew:
30.55
Prob(H) (two-sided):                  0.00   Kurtosis:
1205.31
==============================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
inf. Standard errors may be unstable.
```

2.1.4 Plotted ARIMA Predictions

[21]:
```
best_aic_fcast = best_aic_model.get_prediction(start=train_end, end=len(y_CA) +␣
↪test_horizon - 1)
best_aic_yhat = best_aic_fcast.predicted_mean
best_aic_rmse = np.sqrt(mean_squared_error(ytest_CA, best_aic_yhat.iloc[:
↪len(ytest_CA)]))

best_bic_fcast = best_bic_model.get_prediction(start=train_end, end=len(y_CA) +␣
↪test_horizon -1)
best_bic_yhat = best_bic_fcast.predicted_mean
best_bic_rmse = np.sqrt(mean_squared_error(ytest_CA, best_bic_yhat.iloc[:
↪len(ytest_CA)]))

best_rmse_fcast = best_rmse_model.get_prediction(start=train_end, end=len(y_CA)␣
↪+ test_horizon - 1)
best_rmse_yhat = best_rmse_fcast.predicted_mean
best_rmse_rmse = np.sqrt(mean_squared_error(ytest_CA, best_rmse_yhat.iloc[:
↪len(ytest_CA)]))


plt.figure(figsize=(10,6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA[(train_end - 150):],␣
↪color="steelblue", label="Training Data")
plt.plot(ttest, ytest_CA, color="red", label="Actual Future Values")

plt.plot(tpred, best_aic_yhat, color="blue",
         label=f"Predicted Future Values\nBest AIC ARIMA({best_aic['p']:.0f},␣
↪{best_aic['d']:.0f}, {best_aic['q']:.0f}) Model")
plt.plot(tpred, best_bic_yhat, color="yellow",
         label=f"Predicted Future Values\nBest BIC ARIMA({best_bic['p']:.0f},␣
↪{best_bic['d']:.0f}, {best_bic['q']:.0f}) Model")
plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Predicted Future Values\nBest RMSE ARIMA({best_rmse['p']:.0f},␣
↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model")

plt.axvline(x=train_end, color="grey", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
↪(April 11, 2026)")

plt.xlabel("Time (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.title("Initial Claims Filed in California")
plt.legend(); plt.show();

print(f"Root Mean Squared Error of Best AIC ARIMA({best_aic['p']:.0f},␣
↪{best_aic['d']:.0f}, {best_aic['q']:.0f}) Model: {best_aic_rmse}")
```

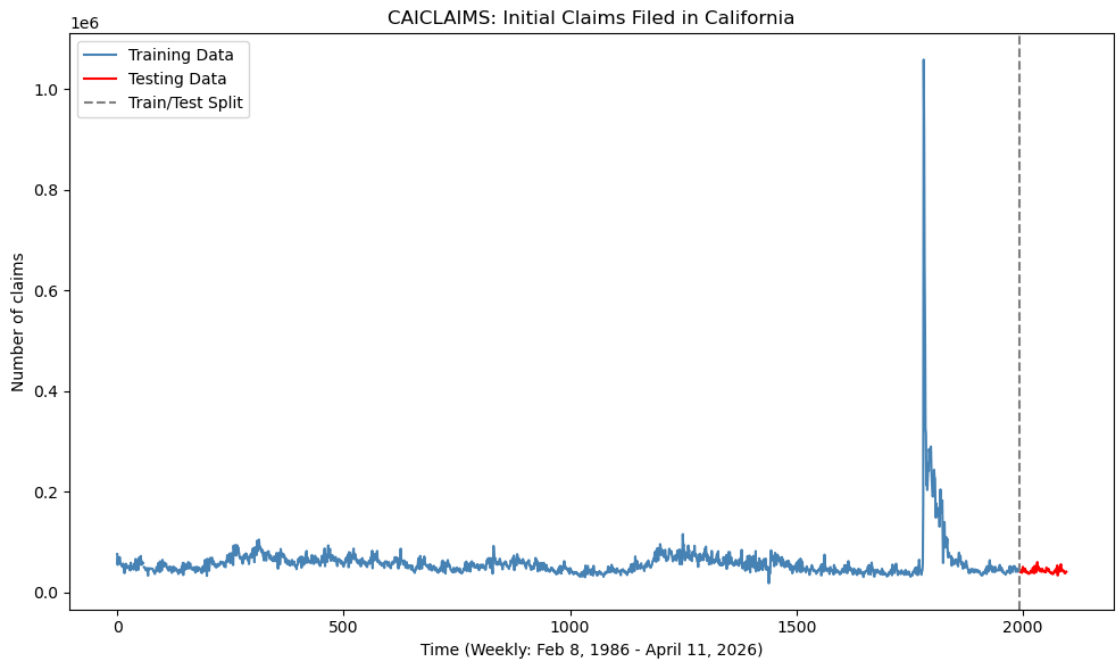
```
print(f"Root Mean Squared Error of Best BIC ARIMA({best_bic['p']:.0f},␣
↪{best_bic['d']:.0f}, {best_bic['q']:.0f}) Model: {best_bic_rmse}")
print(f"Root Mean Squared Error of Best RMSE ARIMA({best_rmse['p']:.0f},␣
↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {best_rmse_rmse}")
```



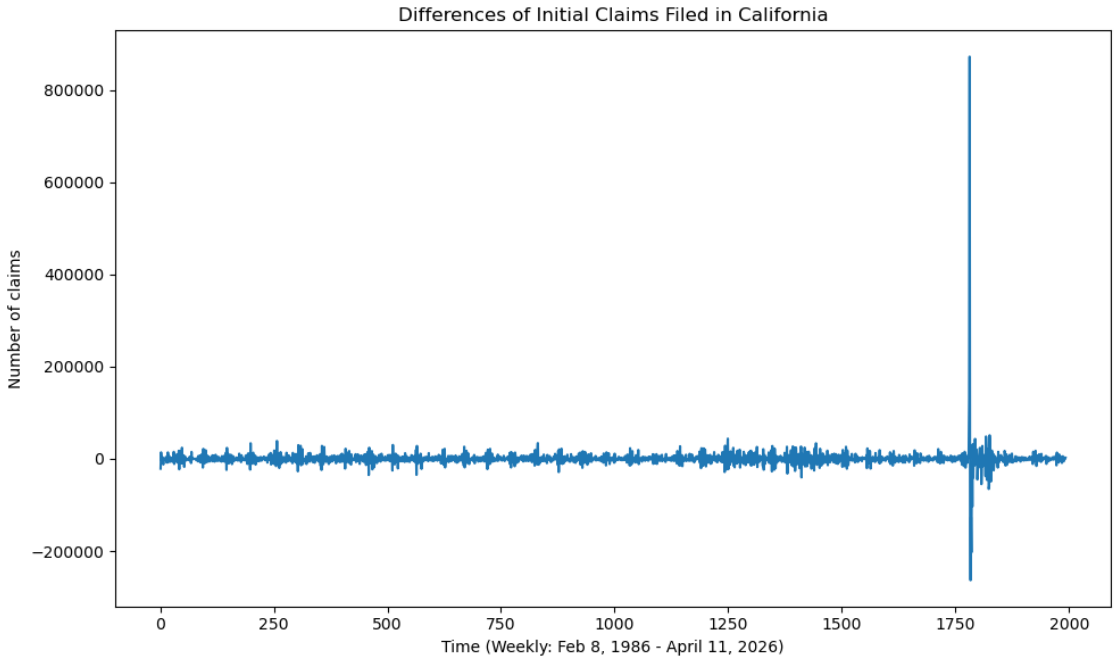

```
Root Mean Squared Error of Best AIC ARIMA(3, 1, 5) Model: 4804.807622002152
Root Mean Squared Error of Best BIC ARIMA(0, 1, 6) Model: 4809.157796229412
Root Mean Squared Error of Best RMSE ARIMA(8, 1, 0) Model: 4598.93312709685
```

2.2 2. SARIMAX with Fourier Frequencies, Before/After COVID Dummy

[15]:
```
plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```



[16]:
```
plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.title("Differences of Initial Claims Filed in California")
plt.tight_layout()
plt.show();
```


Differences of Initial Claims Filed in California
Number of claims
800000
600000
400000
200000
0
−200000
0
250
500
750
1000
1250
1500
1750
2000
Time (Weekly: Feb 8, 1986 - April 11, 2026)

2.2.1 Design Matrix Construction

[17]:
```
def make_features(n, covid_start=1780):
    t = np.arange(n)
    X = pd.DataFrame(index=np.arange(n))

    # linear trend
    X["trend"] = t

    # yearly seasonality for weekly data
    for k in range(1, 4):  # 1st, 2nd, 3rd harmonic
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)

    # COVID level-shift dummy
    X["covid_after"] = (t >= covid_start).astype(int)

    return X
```

2.2.2 SARIMAX Model Test

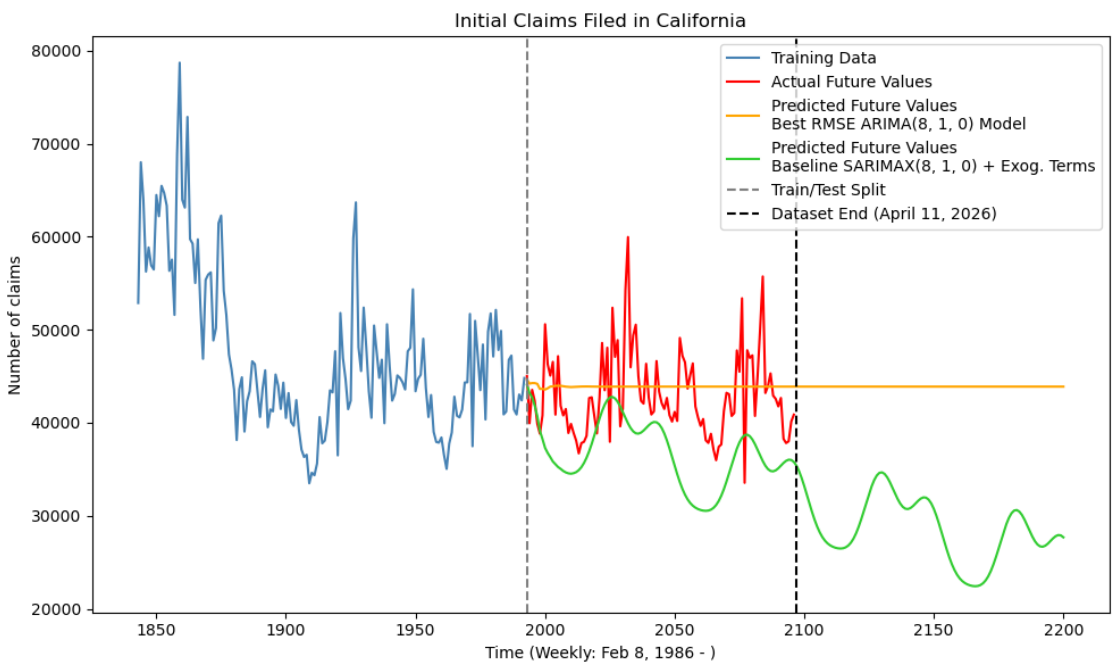
[18]:
```
# Exog features through observed data only (for holdout RMSE / grid search)
X_CA = make_features(len(y_CA), covid_start=1780)

Xtrain_CA = X_CA.iloc[:train_end]
Xtest_CA = X_CA.iloc[train_end:]
Xtest_CA = Xtest_CA[Xtrain_CA.columns]

# Exog through observed data + future period (for extended future forecasts)
X_CA_extended = make_features(len(y_CA) + test_horizon, covid_start=1780)

Xtrain_CA_extended = X_CA_extended.iloc[:train_end]
Xtest_CA_extended = X_CA_extended.iloc[train_end : len(y_CA) + test_horizon]
Xtest_CA_extended = Xtest_CA_extended[Xtrain_CA_extended.columns]


# Best RMSE Baseline ARIMA(8, 1, 0) model with exogenous regressors
sarimax_model = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_extended,
    order=(best_rmse['p'], best_rmse['d'], best_rmse['q']),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

print(sarimax_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(8, 1, 0)   Log Likelihood              -22607.734
Date:                Tue, 28 Apr 2026   AIC                          45249.469
Time:                        18:01:44   BIC                          45344.547
Sample:                             0   HQIC                         45284.394
                               - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
trend         -77.9511    616.206     -0.127      0.899   -1285.693    1129.791
sin_1        2503.6547   7189.390      0.348      0.728   -1.16e+04    1.66e+04
cos_1        3316.2240   7590.046      0.437      0.662   -1.16e+04    1.82e+04
sin_2       -1751.3062   4282.271     -0.409      0.683   -1.01e+04    6641.791
cos_2         233.6257   3987.129      0.059      0.953   -7581.003    8048.254
sin_3        -668.0537   3333.770     -0.200      0.841   -7202.122    5866.015
cos_3         667.4854   3675.067      0.182      0.856   -6535.513    7870.484
covid_after  1.284e+05    2.5e+04      5.137      0.000    7.94e+04    1.77e+05
```

```
ar.L1          -0.1660      0.034     -4.815      0.000      -0.234      -0.098
ar.L2          -0.2215      0.008    -28.494      0.000      -0.237      -0.206
ar.L3          -0.1810      0.014    -13.078      0.000      -0.208      -0.154
ar.L4          -0.2376      0.009    -27.328      0.000      -0.255      -0.221
ar.L5          -0.1206      0.013     -9.239      0.000      -0.146      -0.095
ar.L6          -0.2005      0.008    -23.683      0.000      -0.217      -0.184
ar.L7          -0.0944      0.014     -6.589      0.000      -0.122      -0.066
ar.L8          -0.1297      0.008    -17.012      0.000      -0.145      -0.115
sigma2         4.9e+08    123.318   3.97e+06      0.000     4.9e+08     4.9e+08
================================================================================
===
Ljung-Box (L1) (Q):                   0.17   Jarque-Bera (JB):
127823163.88
Prob(Q):                              0.68   Prob(JB):
0.00
Heteroskedasticity (H):              20.88   Skew:
31.17
Prob(H) (two-sided):                  0.00   Kurtosis:
1244.92
================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
3.46e+20. Standard errors may be unstable.
```

[22]:
```python
# Forecast test period
sarimax_fcast = sarimax_model.get_forecast(steps=len(ytest_CA) + test_horizon,␣
↪exog=Xtest_CA_extended)
sarimax_yhat = sarimax_fcast.predicted_mean
sarimax_yhat.index = tpred
sarimax_rmse = np.sqrt(mean_squared_error(ytest_CA, sarimax_yhat.iloc[:
↪len(ytest_CA)]))


# Plot predictions against ARIMA baselines and actual values
plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Future Values", color="red")

plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Predicted Future Values\nBest RMSE ARIMA({best_rmse['p']:.0f},␣
↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model")
plt.plot(tpred, sarimax_yhat, color="limegreen",
```

```
        label=f"Predicted Future Values\nBaseline SARIMAX({best_rmse['p']:.
↪0f}, {best_rmse['d']:.0f}, {best_rmse['q']:.0f}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split"␣
↪)
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
↪(April 11, 2026)")

plt.title("Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();


print(f"Root Mean Squared Error of Best RMSE ARMIA({best_rmse['p']:.0f},␣
↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {best_rmse_rmse}")
print(f"Room Mean Squared Error of Baseline SARIMAX ({best_rmse['p']:.0f},␣
↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {sarimax_rmse}")
```




```
Root Mean Squared Error of Best RMSE ARMIA(8, 1, 0) Model: 4598.93312709685
Room Mean Squared Error of Baseline SARIMAX (8, 1, 0) Model: 7861.362181423878
```

2.2.3 Parameter Selection (Grid Search)

[23]:
```
pmax, dmax, qmax = 8, 1, 6

def fit_sarimax_exog(params):
    warnings.filterwarnings("ignore")

    p, d, q = params

    try:
        model = SARIMAX(
            ytrain_CA,
            exog=Xtrain_CA,
            order=(p, d, q),
            enforce_stationarity=False,
            enforce_invertibility=False
        ).fit(disp=False)

        forecast = model.get_forecast(
            steps=len(ytest_CA),
            exog=Xtest_CA
        )

        y_hat = forecast.predicted_mean
        y_hat.index = ytest_CA.index

        rmse = np.sqrt(mean_squared_error(ytest_CA, y_hat))

        return {
            "p": p,
            "d": d,
            "q": q,
            "AIC": model.aic,
            "BIC": model.bic,
            "RMSE": rmse
        }

    except Exception:
        return None
```

[24]:
```
cache_path_sarimax = "sarimax_grid_results_CAICLAIMS.csv"

param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))
```

```python
if os.path.exists(cache_path_sarimax):
    print(f"Loading cached SARIMAX grid search results from␣
 ↪{cache_path_sarimax}")
    results_sarimax_df = pd.read_csv(cache_path_sarimax)

else:
    print("Running SARIMAX grid search...")

    results_sarimax = Parallel(n_jobs=-1, verbose=10)(
        delayed(fit_sarimax_exog)(params) for params in param_grid
    )

    results_sarimax = [r for r in results_sarimax if r is not None]
    results_sarimax_df = pd.DataFrame(results_sarimax)

    if results_sarimax_df.empty:
        raise ValueError(
            "SARIMAX grid search failed: no models were successfully fit. "
            "Check that len(Xtest_CA) equals len(ytest_CA)."
        )

    results_sarimax_df.to_csv(cache_path_sarimax, index=False)
    print(f"Saved SARIMAX grid search results to {cache_path_sarimax}")
```

```
Running SARIMAX grid search…

[Parallel(n_jobs=-1)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=-1)]: Done   2 tasks      | elapsed:    0.4s
[Parallel(n_jobs=-1)]: Done   9 tasks      | elapsed:    1.9s
[Parallel(n_jobs=-1)]: Done  16 tasks      | elapsed:    2.9s
[Parallel(n_jobs=-1)]: Done  25 tasks      | elapsed:    6.0s
[Parallel(n_jobs=-1)]: Done  34 tasks      | elapsed:    8.8s
[Parallel(n_jobs=-1)]: Done  45 tasks      | elapsed:   14.5s
[Parallel(n_jobs=-1)]: Done  56 tasks      | elapsed:   26.6s
[Parallel(n_jobs=-1)]: Done  69 tasks      | elapsed:   37.4s
[Parallel(n_jobs=-1)]: Done  82 tasks      | elapsed:   51.1s
[Parallel(n_jobs=-1)]: Done  97 tasks      | elapsed:  1.1min
[Parallel(n_jobs=-1)]: Done 124 out of 126 | elapsed:  1.9min remaining:    1.8s

Saved SARIMAX grid search results to sarimax_grid_results_CAICLAIMS.csv

[Parallel(n_jobs=-1)]: Done 126 out of 126 | elapsed:  1.9min finished
```

2.2.4 Best Models by Evaluation Metrics

[27]:
```
best_aic_sarimax = results_sarimax_df.loc[results_sarimax_df["AIC"].idxmin()]
best_bic_sarimax = results_sarimax_df.loc[results_sarimax_df["BIC"].idxmin()]
best_rmse_sarimax = results_sarimax_df.loc[results_sarimax_df["RMSE"].idxmin()]

print(
    f"Best model by AIC: SARIMAX({int(best_aic_sarimax['p'])}, "
    f"{int(best_aic_sarimax['d'])}, {int(best_aic_sarimax['q'])}), "
    f"AIC: {best_aic_sarimax['AIC']:.3f}, "
    f"BIC: {best_aic_sarimax['BIC']:.3f}, "
    f"RMSE: {best_aic_sarimax['RMSE']:.3f}"
)

print(
    f"Best model by BIC: SARIMAX({int(best_bic_sarimax['p'])}, "
    f"{int(best_bic_sarimax['d'])}, {int(best_bic_sarimax['q'])}), "
    f"AIC: {best_bic_sarimax['AIC']:.3f}, "
    f"BIC: {best_bic_sarimax['BIC']:.3f}, "
    f"RMSE: {best_bic_sarimax['RMSE']:.3f}"
)

print(
    f"Best model by RMSE: SARIMAX({int(best_rmse_sarimax['p'])}, "
    f"{int(best_rmse_sarimax['d'])}, {int(best_rmse_sarimax['q'])}), "
    f"AIC: {best_rmse_sarimax['AIC']:.3f}, "
    f"BIC: {best_rmse_sarimax['BIC']:.3f}, "
    f"RMSE: {best_rmse_sarimax['RMSE']:.3f}"
)
```

```
Best model by AIC: SARIMAX(8, 1, 1), AIC: 45202.391, BIC: 45303.063, RMSE:
7126.754
Best model by BIC: SARIMAX(8, 1, 1), AIC: 45202.391, BIC: 45303.063, RMSE:
7126.754
Best model by RMSE: SARIMAX(8, 1, 1), AIC: 45202.391, BIC: 45303.063, RMSE:
7126.754
```

[28]:
```
best_p = int(best_rmse_sarimax['p'])
best_d = int(best_rmse_sarimax['d'])
best_q = int(best_rmse_sarimax['q'])

best_rmse_sarimax_model = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_extended,
    order=(best_p, best_d, best_q),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)
```

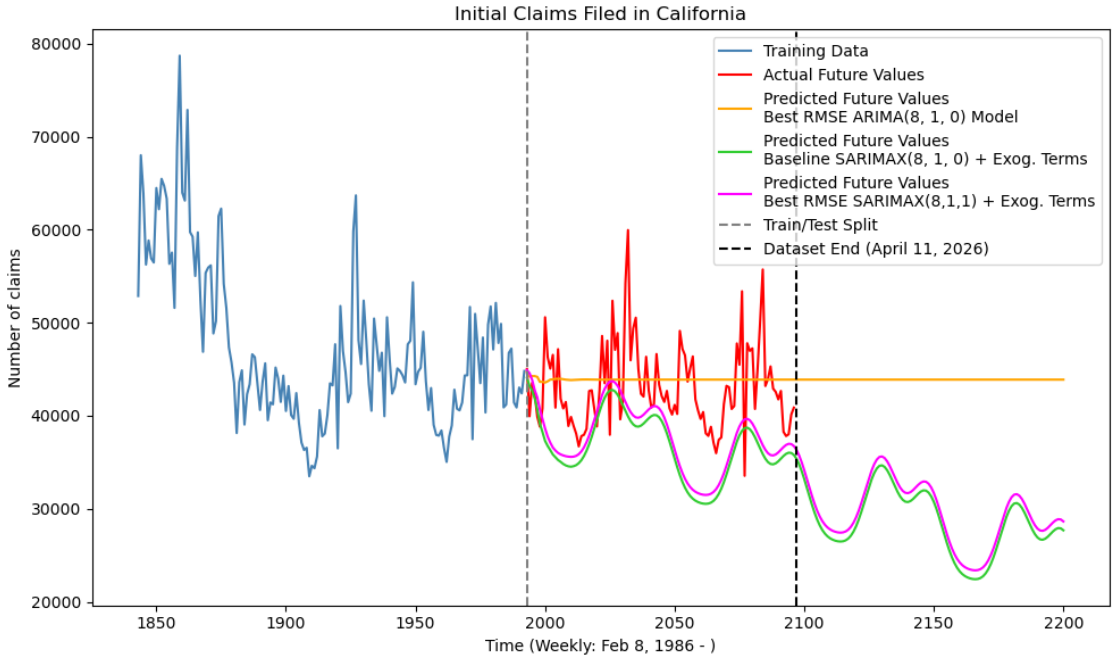
```python
print(best_rmse_sarimax_model.summary())
```

```
                                 SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(8, 1, 1)   Log Likelihood              -22583.196
Date:                Tue, 28 Apr 2026   AIC                          45202.391
Time:                        18:05:44   BIC                          45303.063
Sample:                             0   HQIC                         45239.371
                               - 1993
Covariance Type:                  opg
==================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
trend          -77.9511    378.793     -0.206      0.837    -820.372     664.470
sin_1         2503.6547   5032.125      0.498      0.619   -7359.129    1.24e+04
cos_1         3316.2240   5527.934      0.600      0.549   -7518.327    1.42e+04
sin_2        -1751.3062   4326.977     -0.405      0.686   -1.02e+04    6729.413
cos_2          233.6257   4586.437      0.051      0.959   -8755.625    9222.877
sin_3         -668.0537   4559.327     -0.147      0.884   -9604.171    8268.064
cos_3          667.4854   4687.740      0.142      0.887   -8520.316    9855.286
covid_after   1.284e+05   2.08e+04      6.180      0.000    8.77e+04    1.69e+05
ar.L1            0.6382      0.040     15.772      0.000       0.559       0.718
ar.L2           -0.0935      0.015     -6.061      0.000      -0.124      -0.063
ar.L3           -0.0200      0.063     -0.316      0.752      -0.144       0.104
ar.L4           -0.0997      0.046     -2.177      0.030      -0.190      -0.010
ar.L5            0.0506      0.046      1.105      0.269      -0.039       0.140
ar.L6           -0.1185      0.035     -3.402      0.001      -0.187      -0.050
ar.L7            0.0479      0.056      0.861      0.389      -0.061       0.157
ar.L8           -0.0570      0.048     -1.186      0.235      -0.151       0.037
ma.L1           -0.8380      0.038    -22.250      0.000      -0.912      -0.764
sigma2        4.902e+08     84.745   5.78e+06      0.000     4.9e+08     4.9e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):
141551186.17
Prob(Q):                              0.93   Prob(JB):
0.00
Heteroskedasticity (H):              21.27   Skew:
32.50
Prob(H) (two-sided):                  0.00   Kurtosis:
1309.94
===================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
```

```
step).
[2] Covariance matrix is singular or near-singular, with condition number
3.87e+20. Standard errors may be unstable.
```

2.2.5 Plotted SARIMAX Predictions

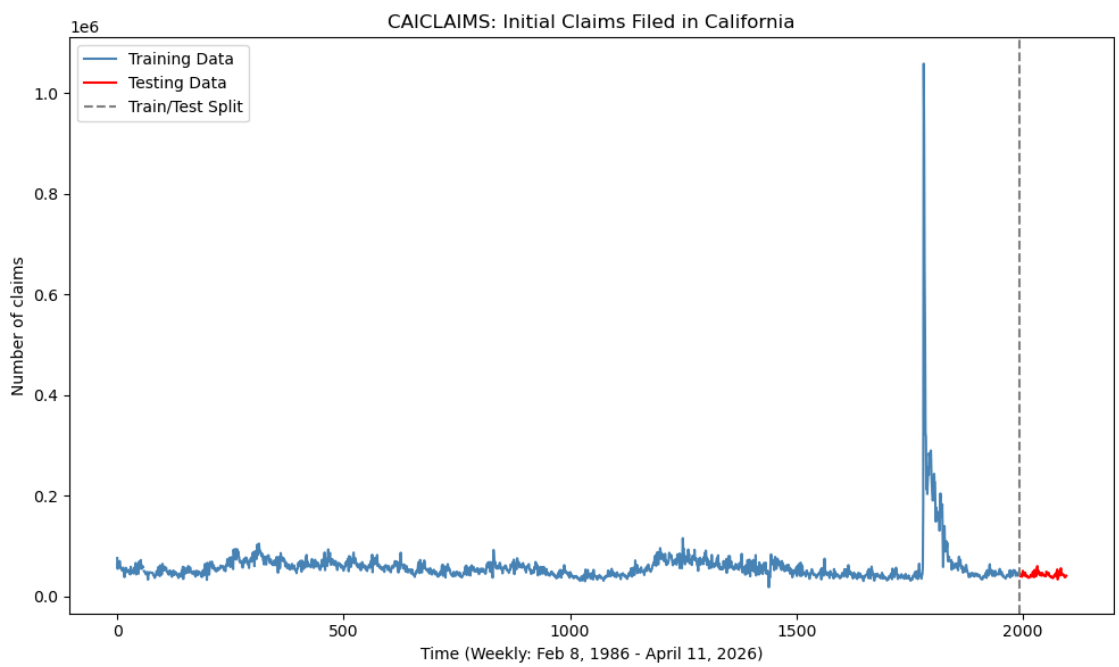
[30]:
```
best_rmse_sarimax_fcast = best_rmse_sarimax_model.
 ↪get_forecast(steps=len(ytest_CA) + test_horizon, exog=Xtest_CA_extended)

best_rmse_sarimax_yhat = best_rmse_sarimax_fcast.predicted_mean
best_rmse_sarimax_yhat.index = tpred
best_sarimax_rmse = np.sqrt(mean_squared_error(ytest_CA, best_rmse_sarimax_yhat.
 ↪iloc[:len(ytest_CA)]))


plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Future Values", color="red")

plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Predicted Future Values\nBest RMSE ARIMA({best_rmse['p']:.0f},␣
 ↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model")
plt.plot(tpred, sarimax_yhat, color="limegreen",
         label=f"Predicted Future Values\nBaseline SARIMAX({best_rmse['p']:.
 ↪0f}, {best_rmse['d']:.0f}, {best_rmse['q']:.0f}) + Exog. Terms")
plt.plot(tpred, best_rmse_sarimax_yhat, color="magenta",
         label=f"Predicted Future Values\nBest RMSE␣
 ↪SARIMAX({best_p},{best_d},{best_q}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")

plt.title("Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();


print(f"Root Mean Squared Error of Best RMSE ARMIA({best_rmse['p']:.0f},␣
 ↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {best_rmse_rmse}")
print(f"Room Mean Squared Error of Baseline SARIMAX ({best_rmse['p']:.0f},␣
 ↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {sarimax_rmse}")
print(f"Root Mean Squared Error of Best RMSE SARIMAX({best_p:.0f}, {best_d:.
 ↪0f}, {best_q:.0f}) Model: {best_sarimax_rmse}")
```



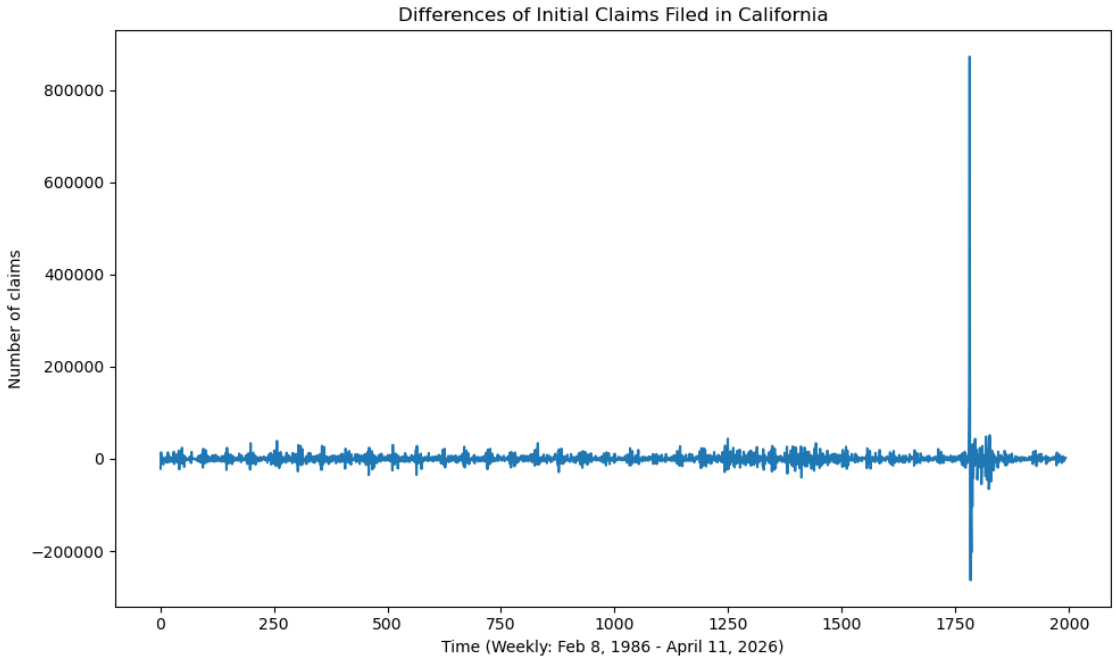

```
Root Mean Squared Error of Best RMSE ARMIA(8, 1, 0) Model: 4598.93312709685
Room Mean Squared Error of Baseline SARIMAX (8, 1, 0) Model: 7861.362181423878
Root Mean Squared Error of Best RMSE SARIMAX(8, 1, 1) Model: 7126.754229243658
```

2.3 3. SARIMAX with Fourier Frequencies, During COVID Dummy

[31]:
```
plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```



[32]:
```
plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.title("Differences of Initial Claims Filed in California")
plt.tight_layout()
plt.show();
```


Differences of Initial Claims Filed in California
Number of claims
800000
600000
400000
200000
0
−200000
0
250
500
750
1000
1250
1500
1750
2000
Time (Weekly: Feb 8, 1986 - April 11, 2026)

2.3.1 Design Matrix Construction

[33]:
```
def make_features_new(n, covid_start=1775, covid_end=1860, n_harmonics=3):
    t = np.arange(n)
    X = pd.DataFrame(index=np.arange(n))

    # Linear trend
    X["trend"] = t

    # Weekly data: yearly period is approximately 52 weeks
    for k in range(1, n_harmonics + 1):
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)

    # COVID period indicator
    X["during_covid"] = ((t >= covid_start) & (t <= covid_end)).astype(int)

    return X
```

[34]:
```
# Exog features through observed data only (for holdout RMSE / grid search)
X_CA_new = make_features_new(n=len(y_CA), covid_start=1775, covid_end=1860,␣
 ↪n_harmonics=3)

Xtrain_CA_new = X_CA_new.iloc[:train_end]
Xtest_CA_new = X_CA_new.iloc[train_end : len(y_CA) + test_horizon]
Xtest_CA_new = Xtest_CA_new[Xtrain_CA_new.columns]

# Exog through observed data + future period (for extended future forecasts)
X_CA_new_extended = make_features_new(n=len(y_CA) + test_horizon,␣
 ↪covid_start=1775, covid_end=1860, n_harmonics=3)

Xtrain_CA_new_extended = X_CA_new_extended.iloc[:train_end]
Xtest_CA_new_extended = X_CA_new_extended.iloc[train_end : len(y_CA) +␣
 ↪test_horizon]
Xtest_CA_new_extended = Xtest_CA_new_extended[Xtrain_CA_new_extended.columns]
```

2.3.2 Parameter Selection (Grid Search)

[35]:
```
def rolling_cv_sarimax(
    y,
    X,
    order,
    initial_train_size=1500,
    horizon=52,
    step=26
):
    """
    Rolling-origin cross-validation for SARIMAX.

    At each split:
    - train on y[0:train_end]
    - forecast y[train_end:train_end+horizon]
    - compute RMSE
    """
    y = pd.Series(y).reset_index(drop=True)
    X = X.reset_index(drop=True)

    rows = []

    split_points = range(initial_train_size, len(y) - horizon + 1, step)

    for train_end in split_points:
        test_start = train_end
        test_end = train_end + horizon - 1

        y_train = y.iloc[:train_end]
        y_test = y.iloc[test_start:test_start + horizon]

        X_train = X.iloc[:train_end]
        X_test = X.iloc[test_start:test_start + horizon]

        # Make sure columns match exactly
        X_test = X_test[X_train.columns]

        try:
            model = SARIMAX(
                y_train,
                exog=X_train,
                order=order,
                enforce_stationarity=False,
                enforce_invertibility=False
            ).fit(disp=False)
```

```
        forecast = model.get_forecast(
            steps=horizon,
            exog=X_test
        )

        y_hat = forecast.predicted_mean
        y_hat.index = y_test.index

        rmse = np.sqrt(mean_squared_error(y_test, y_hat))

        rows.append({
            "train_end": train_end,
            "test_start": test_start,
            "test_end": test_end,
            "p": order[0],
            "d": order[1],
            "q": order[2],
            "RMSE": rmse,
            "AIC": model.aic,
            "BIC": model.bic,
            "status": "success"
        })

    except Exception as e:
        rows.append({
            "train_end": train_end,
            "test_start": test_start,
            "test_end": test_end,
            "p": order[0],
            "d": order[1],
            "q": order[2],
            "RMSE": np.nan,
            "AIC": np.nan,
            "BIC": np.nan,
            "status": "failed"
        })

cv_results = pd.DataFrame(rows)

mean_rmse = cv_results["RMSE"].mean(skipna=True)
mean_aic = cv_results["AIC"].mean(skipna=True)
mean_bic = cv_results["BIC"].mean(skipna=True)
n_success = cv_results["RMSE"].notna().sum()

return {
    "p": order[0],
    "d": order[1],
```

```
        "q": order[2],
        "CV_RMSE": mean_rmse,
        "CV_AIC": mean_aic,
        "CV_BIC": mean_bic,
        "n_success": n_success,
        "n_splits": len(cv_results)
    }
```

[36]:
```
cache_path = "sarimax_cv_grid_results_CAICLAIMS.csv"

pmax, dmax, qmax = 8, 1, 6

param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

initial_train_size = 1500
horizon = 52
step = 26


def evaluate_order_cv(order):
    return rolling_cv_sarimax(
        y=ytrain_CA,
        X=Xtrain_CA_new,
        order=order,
        initial_train_size=initial_train_size,
        horizon=horizon,
        step=step
    )


if os.path.exists(cache_path):
    print(f"Loading cached grid search results from {cache_path}")
    results_sarimax_new_df = pd.read_csv(cache_path)

else:
    print("Running SARIMAX rolling-CV grid search...")

    results_sarimax_new = Parallel(n_jobs=-1, verbose=10)(
        delayed(evaluate_order_cv)(order) for order in param_grid
    )

    results_sarimax_new_df = pd.DataFrame(results_sarimax_new)
```

```python
    # Save cache
    results_sarimax_new_df.to_csv(cache_path, index=False)
    print(f"Saved grid search results to {cache_path}")
```

```
Loading cached grid search results from sarimax_cv_grid_results_CAICLAIMS.csv
```

2.3.3 Best Models by Evaluation Metrics

[37]:
```
# Clean and sort results
results_sarimax_new_df = results_sarimax_new_df.dropna(subset=["CV_RMSE"])
results_sarimax_new_df = results_sarimax_new_df.sort_values("CV_RMSE").
 ↪reset_index(drop=True)

print("Top SARIMAX models by rolling-origin CV RMSE:")
display(results_sarimax_new_df.head(10))
```

```
Top SARIMAX models by rolling-origin CV RMSE:
```

	p	d	q	CV_RMSE	CV_AIC	CV_BIC	n_success	n_splits
0	0	1	1	47721.191698	36487.862324	36542.247482	17	17
1	1	1	0	47926.816861	36735.036084	36789.427138	17	17
2	0	1	0	48223.247108	47973.689881	48022.641831	17	17
3	0	1	5	48328.247812	36323.282202	36399.388355	17	17
4	2	1	3	49011.995647	36370.451089	36446.573786	17	17
5	2	1	6	49366.557518	36279.757298	36372.161858	17	17
6	2	1	5	49519.955258	36301.148969	36388.127429	17	17
7	1	1	3	49558.867752	36366.239568	36436.924930	17	17
8	1	1	2	49656.046398	36396.732530	36461.987640	17	17
9	1	1	6	49682.865084	36297.877555	36384.846553	17	17

[38]:
```
best_rmse_sarimax_new = results_sarimax_new_df.iloc[0]

best_p_new = int(best_rmse_sarimax_new['p'])
best_d_new = int(best_rmse_sarimax_new['d'])
best_q_new = int(best_rmse_sarimax_new['q'])

print(f"Best SARIMAX by rolling CV: SARIMAX({best_p_new}, {best_d_new},␣
 ↪{best_q_new})")
print(f"CV RMSE: {best_rmse_sarimax_new['CV_RMSE']:.3f}")
print(f"CV AIC: {best_rmse_sarimax_new['CV_AIC']:.3f}")
print(f"CV BIC: {best_rmse_sarimax_new['CV_BIC']:.3f}")
print(f"Successful CV splits: {int(best_rmse_sarimax_new['n_success'])} /␣
 ↪{int(best_rmse_sarimax_new['n_splits'])}")
```

```
Best SARIMAX by rolling CV: SARIMAX(0, 1, 1)
CV RMSE: 47721.192
CV AIC: 36487.862
CV BIC: 36542.247
Successful CV splits: 17 / 17
```

2.3.4 Final Holdout Evaluation

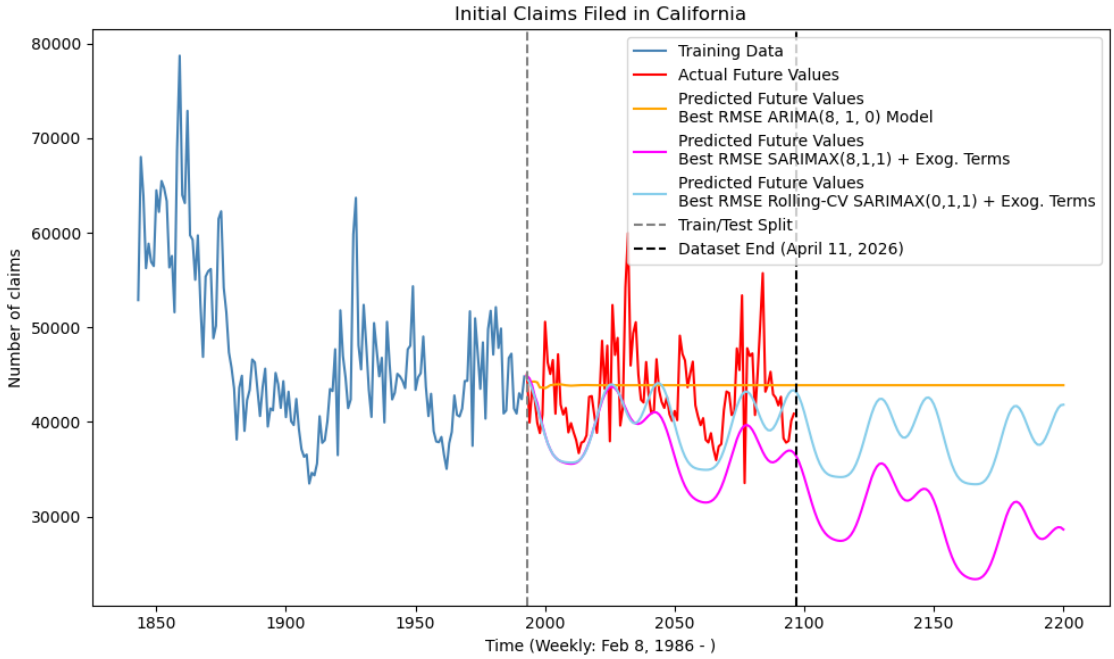
[39]:
```
best_rmse_sarimax_model_new = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_new_extended,
    order=(best_p_new, best_d_new, best_q_new),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

print(best_rmse_sarimax_model_new.summary())


best_rmse_sarimax_fcast_new = best_rmse_sarimax_model_new.get_forecast(
    steps=len(ytest_CA) + test_horizon,
    exog=Xtest_CA_new_extended
)

best_rmse_sarimax_yhat_new = best_rmse_sarimax_fcast_new.predicted_mean
best_rmse_sarimax_yhat_new.index = tpred
best_sarimax_rmse_new = np.sqrt(mean_squared_error(ytest_CA,␣
 ↪best_rmse_sarimax_yhat_new.iloc[:len(ytest_CA)]))

print(f"\nFinal 2-year holdout RMSE for Best RMSE Rolling-CV␣
 ↪SARIMAX({best_p_new:.0f}, {best_d_new:.0f}, {best_q_new:.0f}) Model:␣
 ↪{best_sarimax_rmse_new}")
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(0, 1, 1)   Log Likelihood              -22801.817
Date:                Tue, 28 Apr 2026   AIC                          45623.634
Time:                        18:07:48   BIC                          45679.592
Sample:                             0   HQIC                         45644.186
                               - 1993
Covariance Type:                  opg
==================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
trend          -14.6707   1332.671     -0.011      0.991   -2626.658    2597.316
sin_1         2688.6486   1.67e+04      0.161      0.872   -3.01e+04    3.54e+04
cos_1         2317.0014   1.48e+04      0.157      0.875   -2.66e+04    3.13e+04
sin_2        -2253.6727   6959.108     -0.324      0.746   -1.59e+04    1.14e+04
cos_2           57.8418   7594.943      0.008      0.994   -1.48e+04    1.49e+04
sin_3         -849.1924   4615.611     -0.184      0.854   -9895.624    8197.239
cos_3          970.4282   4498.862      0.216      0.829   -7847.180    9788.037
during_covid  2311.3872      1.440   1605.395      0.000    2308.565    2314.209
ma.L1            0.0282      0.004      7.507      0.000       0.021       0.036
```

```
sigma2        5.567e+08     30.295   1.84e+07      0.000    5.57e+08    5.57e+08
==============================================================================
===
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):
73126243.82
Prob(Q):                              0.90   Prob(JB):
0.00
Heteroskedasticity (H):              21.98   Skew:
24.00
Prob(H) (two-sided):                  0.00   Kurtosis:
940.88
==============================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
4.17e+23. Standard errors may be unstable.

Final 2-year holdout RMSE for Best RMSE Rolling-CV SARIMAX(0, 1, 1) Model:
5810.691323773642
```

2.3.5 Plotted SARIMAX Predictions

[41]:
```
plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Future Values", color="red")

plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Predicted Future Values\nBest RMSE ARIMA({best_rmse['p']:.0f},␣
 ↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model")
plt.plot(tpred, best_rmse_sarimax_yhat, color="magenta",
         label=f"Predicted Future Values\nBest RMSE␣
 ↪SARIMAX({best_p},{best_d},{best_q}) + Exog. Terms")
plt.plot(tpred, best_rmse_sarimax_yhat_new, color="skyblue",
         label=f"Predicted Future Values\nBest RMSE Rolling-CV␣
 ↪SARIMAX({best_p_new},{best_d_new},{best_q_new}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")

plt.title("Initial Claims Filed in California")
plt.xlabel("Time (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();


print(f"Root Mean Squared Error of Best RMSE ARMIA({best_rmse['p']:.0f},␣
 ↪{best_rmse['d']:.0f}, {best_rmse['q']:.0f}) Model: {best_rmse_rmse}")
print(f"Root Mean Squared Error of Best RMSE SARIMAX({best_p:.0f}, {best_d:.
 ↪0f}, {best_q:.0f}) Model: {best_sarimax_rmse}")
print(f"Root Mean Squared Error of Best RMSE Rolling-CV SARIMAX({best_p_new:.
 ↪0f}, {best_d_new:.0f}, {best_q_new:.0f}) Model: {best_sarimax_rmse_new}")
```




```
Root Mean Squared Error of Best RMSE ARMIA(8, 1, 0) Model: 4598.93312709685
Root Mean Squared Error of Best RMSE SARIMAX(8, 1, 1) Model: 7126.754229243658
Root Mean Squared Error of Best RMSE Rolling-CV SARIMAX(0, 1, 1) Model:
5810.691323773642
```